

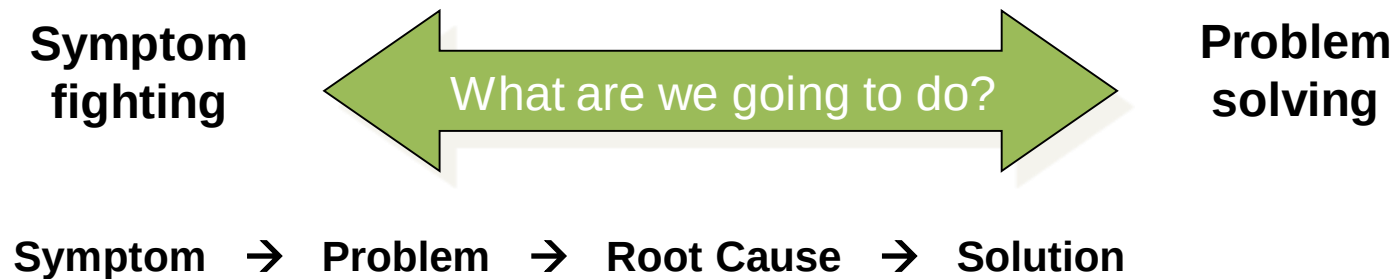
Root Cause Analysis

U5 – Analyse

E3 – Analytical Methods

The element 'Analytical Methods' describes the tools that can be used for risk analysis, root cause analysis and waste identification.

Problem Solving



Symptom

- A symptom is a characteristic or complaint belonging to a specific problem

Problem

- A deviation of expectation; an unexpected effect with unknown causes

Root Cause Analysis

Symptoms and causes

Nurse gives wrong dosage



Doctors writing is illegible



Bloods tests lost



Laboratory has no storage system



Patient late for scan



Wheel chair not in its location



Root Cause Analysis

8D Problem solving



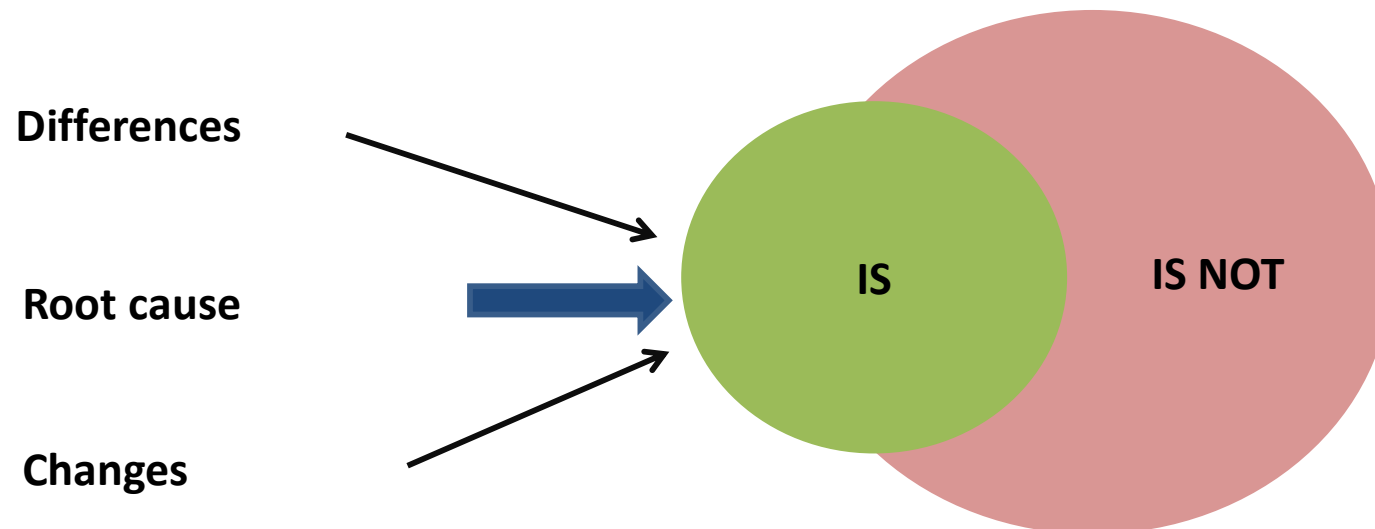
Root Cause Analysis

Describe the problem

Describe the problem as objectively and clearly as possible:

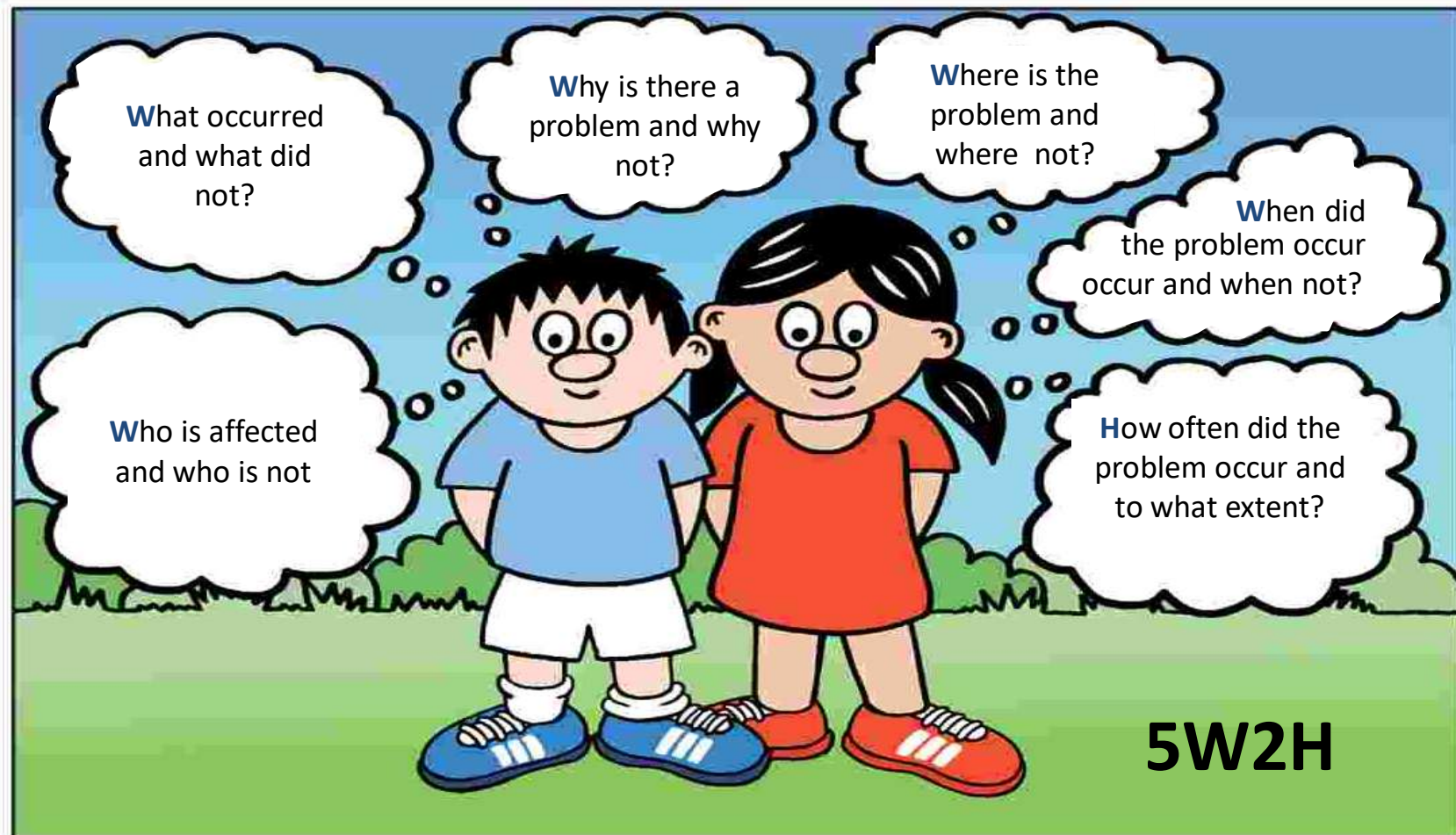
- What IS
- What IS NOT

IS - IS NOT you consciously think about the problem, specifically, the boundaries of what the problem IS or IS NOT



Root Cause Analysis

Is – Is not



Is – Is not

Comparison Analysis

- Differences: What is special, unique or different in terms of what IS vs. what IS NOT?
- Changes: In relation to the differences, what has changed?

Possible Causes

- Events that explain how the problem is caused
- Which changes could have caused the problem?
- Be creative and use if necessary “brainstorming”

Brainstorming

Brainstorming is a group creativity technique designed to quickly generate a large number of ideas for the solution of a problem

Ground rules:

- **Withhold criticism**
- **Focus on quantity**
- **Think 'outside-the-box'**

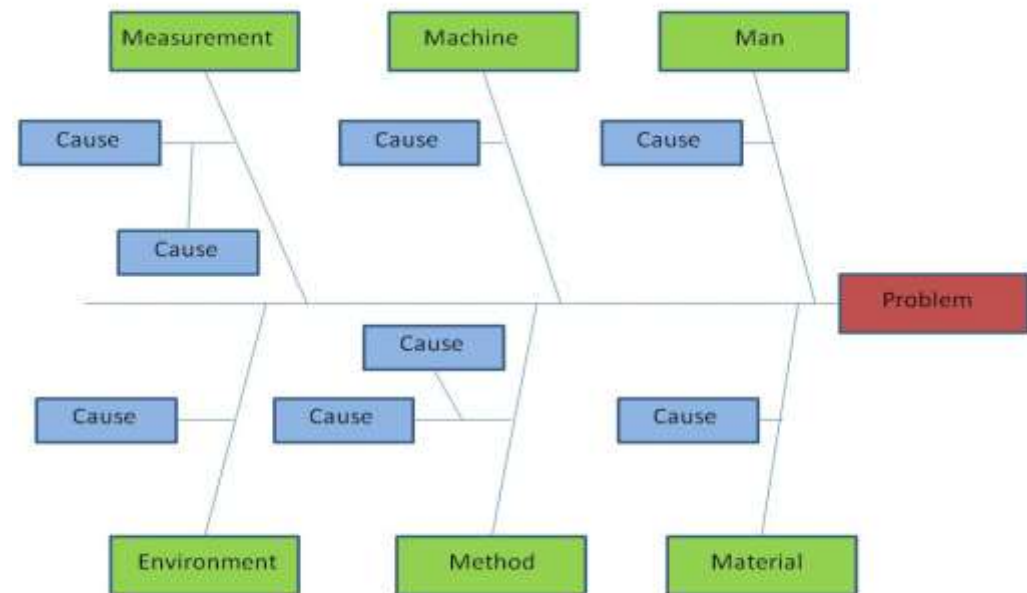


Root Cause Analysis

Cause & effect diagram

Also called:

- Fishbone diagram
- Ishikawa diagram

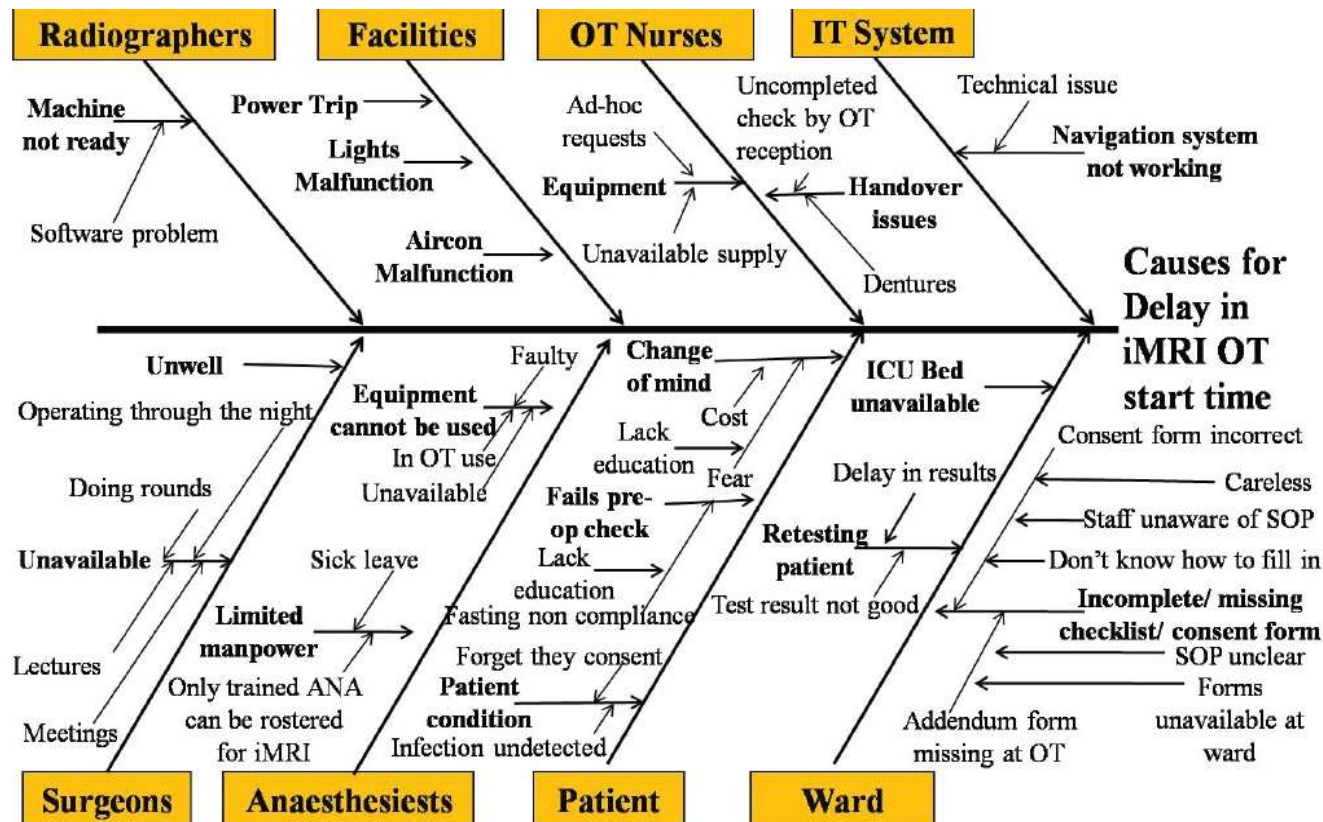


- This diagram clearly displays all factors that contribute to a problem
- It helps to identify relationships between possible causes and problems and to identify data that needs to be collected and analyzed

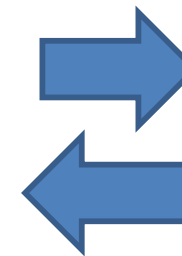
Root Cause Analysis

Example Cause & effect diagram

A C&E diagram used to brainstorm reasons for delays in MRI scan starting times



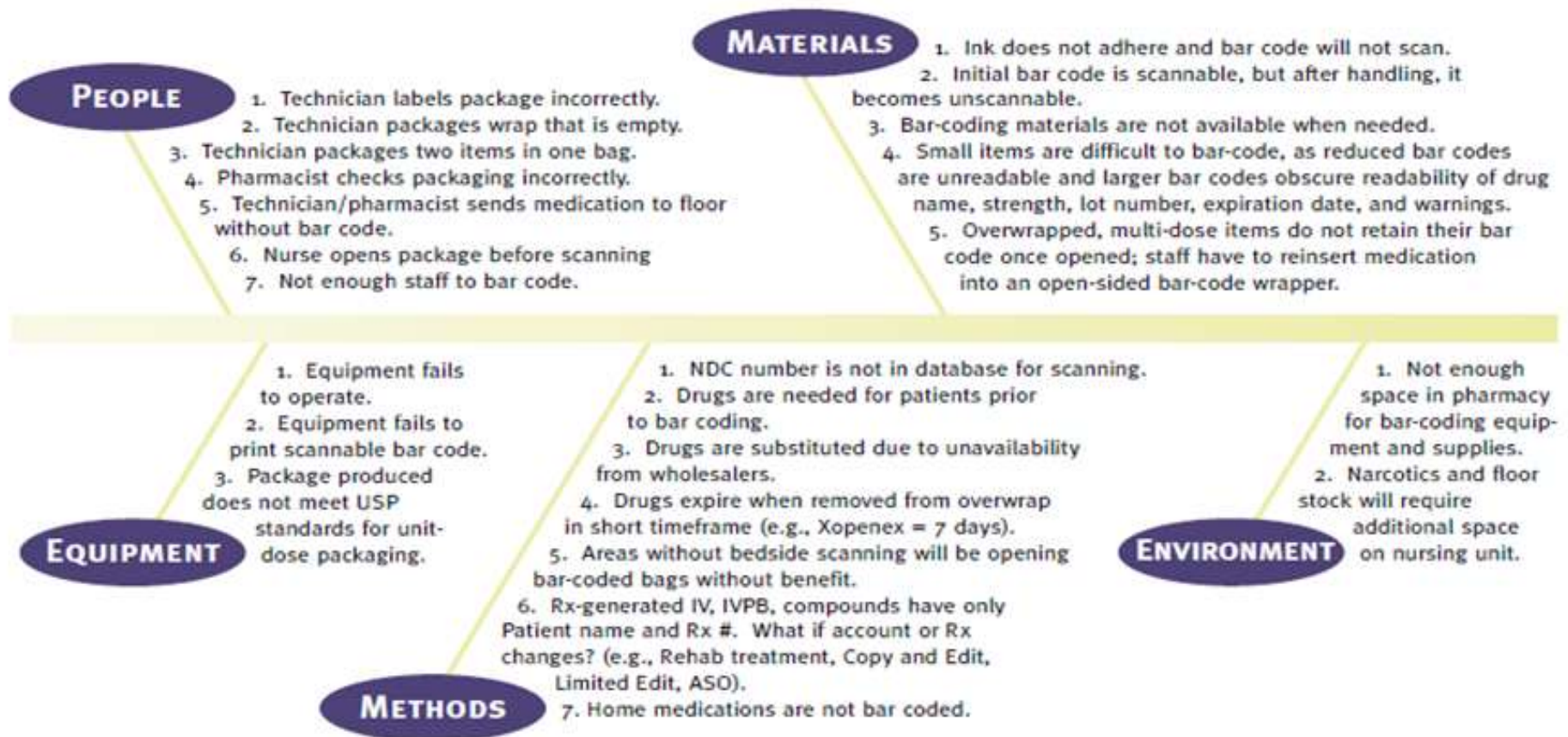
123RF Stock Photo



Root Cause Analysis

Example Cause & effect diagram

Failure to produce accurate, scannable bar codes on medical medications



The 5-Why

- The 5-Why method is not an individual tool, it is part of the whole problem solving toolbox
- It is similar to the Fishbone brainstorming tool

The 5-Why has to answer three questions:

- Why did the problem occur?
- Why was it only detected at the customer?
- Why did “the system” not function?



Example 5-Why: Patient late for therapy

Why was the patient late at the therapy session in the rehabilitation centre, causing delays?

- There was a long wait for a wheelchair

Why was there was a long wait for a trolley?

- Because a replacement wheelchair had to be found

Why did a replacement trolley have to be found?

- Because the original wheelchair was worn and had eventually broken

Why was the original trolley's safety rail worn and broken?

- Because it had not been regularly checked for wear

Why had it not been checked regularly for wear?

- Because there is no equipment maintenance schedule



Example 5-Why: Patient late for therapy

Solution

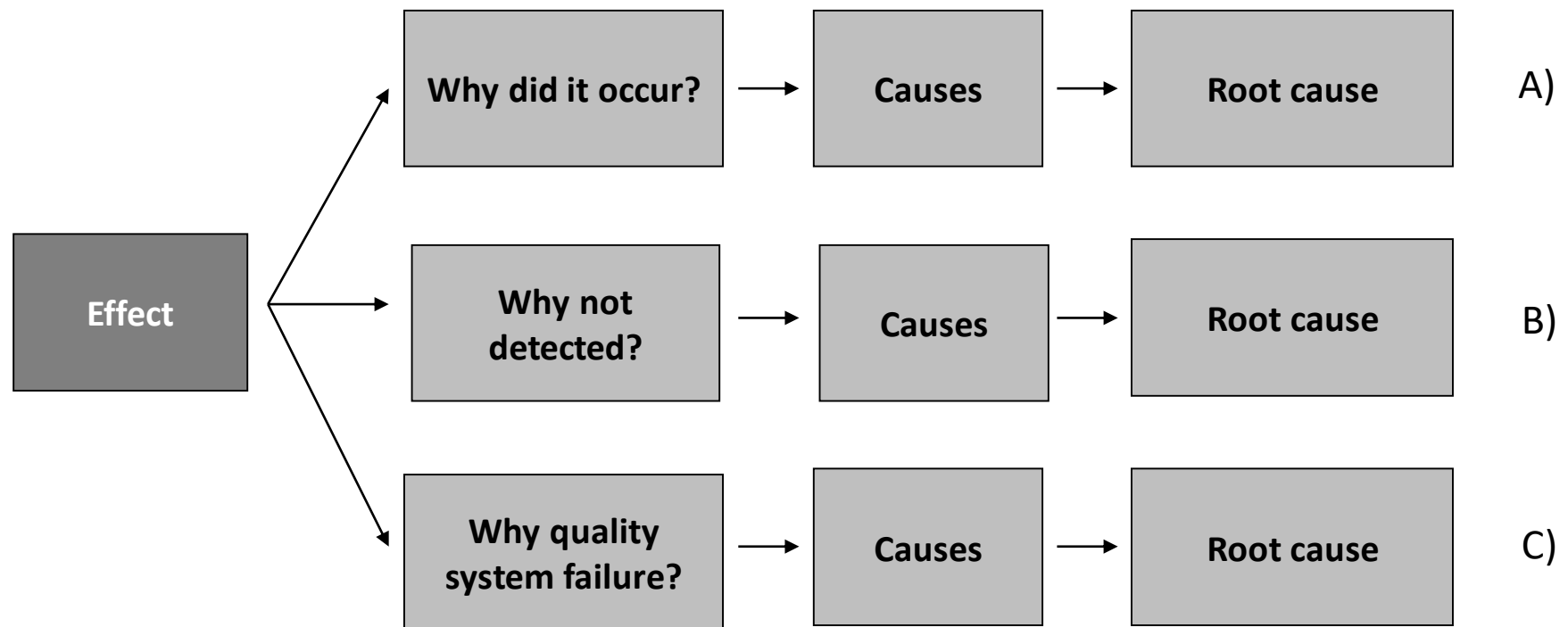
Setting up a proper maintenance schedule helps ensure that patients should never again be late due to faulty equipment.

This reduces delays and improves flow. If you simply repair the wheelchair, the problem may happen again sometime in the future.



The 5-Why

- Use 5-Why technique (ask 5 X WHY)



Example 5-Why: Patient late for diagnostic test

Why are patients arriving late? The porters/ transporters are not called on time*

Why?

- Nurses are busy and often forget
- There is no written request
- No protocol for contacting porters/transport

Why are patients arriving late? Porter is unable to locate patients*

Why?

- The contact request does not include patient location (name only)
- No standard protocol for porter paging

Why are patients arriving late? Patient not ready for transport*

Why?

- Nurses unaware of prescribed test
- No mechanism to inform RN of scheduled procedure

*** Indicates there may be more than one root causes of the problem!**

Proposed solution

The team investigating delayed transport of patients to diagnostic departments discovered that the root cause was:

- **A lack of clear protocol for communicating between the diagnostic department, RN of the clinical department and the transporter/ porter**

Proposed countermeasures:

- **A new protocol where the diagnostic department pages the charge RN and the transporter at the same time (thus eliminating the nurse as the intermediary)**
- **Specified information content of pages**
- **New patient prep procedures involving both the RN and the transporter/ porter**